Atrideb Chatterjee

Contact Information

Kapteyn Astronomical Institute, University of Groningen
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Education

- **Marie Curie Postdoctoral Fellow** April, 2025 - present
  Kapteyn Astronomical Institute, University of Groningen, The Netherlands
- **Postdoctoral Fellow** Dec, 2024 - Mar, 2025
  Kapteyn Astronomical Institute, University of Groningen, The Netherlands
- **Postdoctoral Fellow** Aug, 2022 - Nov, 2024
  Inter-University Centre for Astronomy and Astrophysics, Pune, India
- **Integrated PhD (M.Sc.+Ph.D.)** July, 2015 - July, 2022
  National Centre for Radio Astrophysics, Pune, India
  Supervisor: Prof. Tirthankar Roy Choudhury
- **B.Sc. in Physics** 2012 - 2015
  Scottish Church College, Kolkata, India

Research Interest

- 21-cm Cosmology
- Epoch of Reionization
- Machine Learning
- N-body Simulation

Publications

1. "Impact of Primordial Black Hole population on 21 cm observables at high redshift", **A. Chatterjee**, Barun Maity, Koushiki, 2026, arXiv:2604.21666, submitted to A & A
2. "Effect of Primordial Black Holes on the global 21-cm signal", **A. Chatterjee**, 2026, arXiv:2601.10304, submitted to A & A

3. "Learning Cosmology from Nearest Neighbour Statistics" **A. Chatterjee**, Arka Banerjee, Francisco Villaescusa-Navarro, Tom Abel, 2025, arXiv:2511.13393, accepted in A&A

4. "Exploring the BSM parameter space with Neural Network aided Simulation-Based Inference" **A. Chatterjee**, Arghya Choudhury, Sourav Mitra, Arpita Mondal, Subhadeep Mondal, Journal of High Energy Physics 2025 (12), 138

5. "Cosmology from Point Clouds with Dark Matter Halos from the Quijote Simulations", **A. Chatterjee**, Francisco Villaescusa-Navarro, 2025, The Astrophysical Journal 985 (1), 132

6. "Ruling out Strongly Interacting Dark Matter-Dark Radiation Models from Joint Cosmic Microwave Background-Quasar Observations", **A. Chatterjee**, Sourav Mitra, Amrita Banerjee, 2023, MNRAS Letters, 528 (1), L168-L172

7. "Predictions of the 21cm global signal in the JWST and ALMA era", **A. Chatterjee**, Pratika Dayal, Valentin Mauerhofer, 2023, MNRAS, 525 (1), 620-625

8. "Warm dark Matter constraints from the joint analysis of CMB, Lyman-α, and global 21 cm data", **A. Chatterjee**, Tirthankar Roy Choudhury, 2023, MNRAS, 527 (4), 10777 - 10787

9. "Non-parametric Reconstruction of Photon Escape Fraction from Reionization", Sourav Mitra, **A. Chatterjee**, 2023, MNRAS Letters, 523 (1), L35-L39

10. "Studying Cosmic Dawn using redshifted HI 21-cm signal: A brief review", Ankita Bera, Raghunath Ghara, **A. Chatterjee**, Kanan K. Datta, Saumyadip Samui, 2023, Journal of Astrophysics and Astronomy, 44, 10

11. "CosmoReionMC: A package for estimating cosmological and astrophysical parameters using CMB, Lyman-α absorption and global 21 cm data", **A. Chatterjee**, Tirthankar Roy Choudhury, Sourav Mitra, 2021, MNRAS, 507 (2), 2405-2422

12. "Using Artificial Neural Networks to extract the 21-cm Global Signal from the EDGES data", M. Choudhury, **A. Chatterjee**, A. Datta, Tirthankar Roy Choudhury, 2021, MNRAS, 502 (2), 2815-2825

13. "A hint on the metal-free star formation rate density from 21-cm-EDGES data", **A. Chatterjee**, Pratika Dayal, Tirthankar Roy Choudhury, Raffaella Schneider, 2020, MNRAS, 496 (2), 1445-1452

14. "Ruling out 3 keV warm dark matter using 21-cm EDGES data", **A. Chatterjee**, Pratika Dayal, Tirthankar Roy Choudhury & Anne Hutter, 2019, MNRAS, 487 (3), 3560-3567

Selected Seminar and Conference Talks

- **Majorana-Raychaudhuri Seminar** March, 2026
  Invited talk (Online)
- **MPA Cosmology/Galaxy seminar** June, 2024
  Max Planck Institute for Astrophysics, Garching
- **International Conference on Gravitation and Cosmology (ICGC)** December, 2023
  Indian Institute of Technology Guwahati, Guwahati, India
- **Advanced 21-cm Cosmology Workshop** December, 2023
  National Institute of Science Education and Research, India
- **Workshop on 21-cm Cosmology in the Square Kilometre Array Era** October, 2022
  Indian Statistical Institute, Kolkata, India

- **Learning the high-redshift Universe** February, 2022
  Online Conference
- **State of the Universe Seminar** December 2021
  Tata Institute of Fundamental Research
- **4th 21-cm Global Workshop** October, 2021
  University of Colorado Boulder (online conference)
- **EoR India workshop** April, 2021
  Online seminar
- **Seminar** November, 2020
  Inter-University Centre for Astronomy and Astrophysics
- **3rd 21-cm Global Workshop** November, 2020
  Cambridge, UK (online conference)
- **The First Billion Years of the Universe** January, 2020
  Indian Institute of Technology, Indore, India
- **COSMOLOGY - THE NEXT DECADE** January, 2019
  International Centre for Theoretical Science, Bangalore, India
- **International Conference on Gravitation and Cosmology (ICGC)** December, 2019
  Indian Institute of Science Education and Research, Mohali, India
- **Frontiers in 21 cm Cosmology** December, 2018
  Indian Institute of Astrophysics, Bangalore, India

Programming Skills

Python, PyTorch and Fortran

Teaching Experience

- Project guide for Bachelor's research (Astronomy) of Tejaswi Verma April, 2025 - July, 2025
  Kapteyn Astronomical Institute, Faculty of Science and Engineering, University of Groningen
- External project guide for B.Sc. (Physics) thesis of Manila Anil Ramrakhyani
  Fergusson College (Autonomous), Pune December 2022 - May 2023
- Dept. of Physics, Savitribai Phule Pune University, India August - December 2018
  (**Grader** for the M.Sc. course **"Astronomy & Astrophysics"**) August - December 2019

Awards

- Awarded Marie Skodowska-Curie Actions (MSCA) Postdoctoral Fellowship